



Kingdom of Saudi Arabia
Ministry of Higher Education
King Abdulaziz University
Faculty of Computing and Information Technology

CPCS499 — Senior Project

AuditAI: AI-Powered External Auditor

Design and Development of an Intelligent Financial Auditing System for Enterprise External Audit Operations

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Date of Submission: Jun 14, 2026

Declaration of Originality by Students

We hereby declare that this project report is based on our original work except for citations and quotations, which have been duly acknowledged. This report has not been submitted previously for any academic degree or professional qualification at King Abdulaziz University or any other institution.

The AuditAI system — including its audit engine, PDF processing pipeline, Smart Column Detector, AR/AP reconciliation logic, and web application — was entirely designed, implemented, and tested by our team within the scope of CPCS498 and CPCS499 Senior Projects under the supervision of Dr. Asaad Ahmed.

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Acknowledgement

We would like to express our sincere gratitude to our supervisor, Dr. Asaad Ahmed, for his continuous guidance, constructive feedback, and encouragement throughout both phases of this senior project. His expertise in software engineering and his patience during our technical consultations were invaluable in shaping AuditAI into a complete and functional system.

We also extend our thanks to the Faculty of Computing and Information Technology at King Abdulaziz University for providing the academic environment, resources, and structured curriculum that enabled us to tackle a real-world problem in financial auditing.

Special appreciation goes to the company that generously provided their real financial datasets — including approximately 500 PDF invoices, 8 Excel financial files spanning multiple years — which made it possible to test AuditAI against authentic data and produce meaningful, measurable results.

Finally, we thank our families for their unwavering support and patience throughout this journey.

Abstract

Financial reconciliation and validation processes in external auditing are time-consuming, error-prone, and heavily reliant on manual effort. Traditional audit methods struggle to efficiently handle large volumes of financial transactions and often fail to detect subtle inconsistencies across multiple data sources. This project presents AuditAI, an intelligent AI-powered external auditing system designed to automate financial reconciliation and anomaly detection for enterprise-level audit operations.

AuditAI accepts ten standard financial files — including sales and purchase invoice Excel files, receipt and payment files, AR/AP opening and closing balance files, and PDF invoice files. The system performs a comprehensive four-module audit covering: (1) sales invoice validation including VAT consistency and sequential integrity checks, (2) purchase order validation, (3) AR reconciliation using the formula $AR_Open + Sales - Receipts = AR_Close$, and (4) AP reconciliation using $AP_Open + Purchases - Payments = AP_Close$. A hybrid PDF validation module (pdfplumber + Tesseract OCR) additionally cross-checks PDF invoice documents against structured Excel records.

The system was tested using approximately 500 PDF invoices and 8 Excel financial files from a real company. Results showed 0 VAT errors, 0 duplicate invoices, and 0 missing invoice sequences. The PDF extraction success with an overall invoice match rate of 96.6%. The system processes a full audit in 18–32 minutes (approximately 4.9 seconds per invoice) and generates structured outputs including an interactive web dashboard and downloadable Excel reports. The overall audit score for the company's data was 80/100 (medium risk level).

AuditAI successfully transforms the auditing process from manual effort into intelligent automation, reducing human error and significantly improving audit efficiency. Future work includes expanding AI-based anomaly detection capabilities, integrating with cloud-based accounting platforms, and supporting multi-language financial document processing.

المخلص

تُعدّ عمليات المطابقة المالية والتحقق منها في عمليات التدقيق الخارجي من أكثر العمليات استهلاكاً للوقت وعرضةً للأخطاء البشرية، إذ تعتمد اعتماداً كبيراً على الجهد اليدوي. تُقدّم هذه الورقة نظام AuditAI، وهو نظام تدقيق خارجي ذكي يعتمد على الذكاء الاصطناعي، مصمّم لأتمتة عمليات المطابقة المالية واكتشاف الحالات الشاذة على مستوى المؤسسات.

يقبل النظام ثماني ملفات Excel قياسية، ويُجري تدقيقاً شاملاً يغطي أربعة محاور رئيسية: التحقق من فواتير المبيعات (بما يشمل احتساب ضريبة القيمة المضافة والتحقق من التسلسل الرقمي)، والتحقق من أوامر الشراء، ومطابقة حسابات القبض، ومطابقة حسابات الدفع. كما يتضمن النظام وحدة تحقق مستندي تعتمد على تقنية التعرف الضوئي على الحروف (OCR) للتحقق من فواتير PDF مقارنةً بالسجلات المنظمة.

جرى اختبار النظام على ما يقارب 500 فاتورة PDF و8 ملفات Excel من بيانات شركة حقيقية، وأظهرت النتائج نسبة نجاح استخراج مع مطابقة الفواتير 96.6%، مع صفر أخطاء ضريبية أو فواتير مكررة. يُعالج النظام عملية التدقيق الكاملة خلال 18-32 دقيقة، ويُنتج تقارير منظمة ولوحة تحكم تفاعلية، بلغت درجة التدقيق الإجمالية 80/100 لبيانات الشركة (مستوى مخاطر متوسط). يُشكّل AuditAI خطوة نحو تحويل التدقيق من جهد يدوي إلى أتمتة ذكية، مما يُقلل من الأخطاء البشرية ويرفع كفاءة التدقيق المالي بشكل ملحوظ.

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Chapter 1: Introduction

1.1 Introduction

Financial auditing is one of the most critical functions in modern business governance, ensuring that financial records are accurate, complete, and free from intentional or unintentional errors. Traditionally, external auditors spend a significant portion of their time manually reviewing thousands of invoices, reconciling accounts receivable and accounts payable ledgers, and cross-verifying transaction records against bank statements and payment receipts. This manual approach is slow, expensive, and highly susceptible to human error, particularly when dealing with large enterprises that process hundreds or thousands of transactions per month.

AuditAI addresses this challenge by providing an intelligent, AI-powered external auditing platform that automates the most labor-intensive aspects of the audit process. The system ingests standard financial files — including Excel-based invoice records, receipt and payment vouchers, and scanned or digital PDF invoices — and processes them through four specialized audit modules: sales invoice validation, purchase invoice validation, accounts receivable (AR) reconciliation, and accounts payable (AP) reconciliation. A hybrid PDF processing pipeline combining pdfplumber for digital documents and Tesseract OCR for scanned files further enables the system to cross-verify invoice data extracted from PDF documents against structured Excel records, supporting both Arabic and English text.

The primary benefit of AuditAI is the dramatic reduction in audit time and human error. Tasks that would traditionally require days of manual checking can be completed in 18–32 minutes, with results presented through an interactive web dashboard that highlights exceptions, flags discrepancies, and assigns a risk score to the overall financial health of the audited entity. By making the auditing process faster, more accurate, and more accessible, AuditAI creates real value for external auditors, accounting departments, and small-to-medium enterprises seeking affordable and reliable audit tools.

1.2 Project Proposal

1.2.1 Problem Statement

Manual financial auditing of large datasets presents three core challenges. First, it is extremely time-intensive: a single auditor reviewing 500 invoices must manually check each entry for VAT accuracy, sequential numbering integrity, and cross-reference it against bank and ledger records — a process that can take several days. Second, it is error-prone: human auditors are susceptible to fatigue, overlooked entries, and inconsistent application of accounting rules, particularly when datasets span multiple file formats and naming conventions. Third, existing auditing software is either prohibitively expensive for small enterprises or insufficiently intelligent to handle the heterogeneous data formats common in Saudi Arabian businesses, which frequently mix Arabic and English column headers and document structures.

The project scope encompasses the application of automation and rule-based AI logic to financial data analysis. The system is capable of reading data from Excel spreadsheets and PDF invoices, applying validation and reconciliation rules derived from Saudi accounting standards (including 15% VAT compliance), and producing structured audit findings. The target users are external auditors, internal accounting staff, and financial departments at small to medium-sized enterprises.

1.2.2 Suggested Solution

The suggested solution comprises the following components. An invoice validation engine checks all sales and purchase invoices for VAT accuracy (15% of pre-tax amount), sequential numbering integrity (detecting gaps and duplicates), and total consistency. An AR/AP reconciliation engine applies double-entry accounting formulas to verify that opening balances, transaction records, and closing balances are internally consistent. A Smart Column Detector maps Arabic and English column names to standardized internal field names, enabling the system to process financial files from any company regardless of their formatting conventions. A hybrid PDF processing module extracts invoice data from both digital and scanned PDFs using pdfplumber as the primary method and Tesseract OCR as a fallback, supporting Arabic and English text. Finally, a web application provides user authentication, client management, document upload, an interactive results dashboard, and Excel export functionality.

1.3 Project Aims

The following specific and measurable aims guided the development of AuditAI:

1. Develop a virtual external auditor system that validates financial data from multiple formats (Excel, PDF, scanned invoices) using a hybrid extraction pipeline combining pdfplumber and Tesseract OCR.
2. Implement invoice validation logic that automatically detects missing invoice sequence gaps, duplicate invoice numbers, and 15% VAT calculation errors across sales and purchase records.
3. Build AR and AP reconciliation modules that apply the standard accounting formulas ($AR_Open + Sales - Receipts = AR_Close$; $AP_Open + Purchases - Payments = AP_Close$) to verify financial consistency across customer and supplier records.
4. Design a Smart Column Detector capable of automatically recognizing and mapping both Arabic and English column naming conventions, enabling the system to handle diverse company file formats without manual configuration.
5. Create an interactive web-based dashboard that presents audit findings, exception summaries, KPI cards, and risk classifications in a clear and accessible format for professional auditors.
6. Provide a quantitative risk classification system that assigns an overall audit score (0–100) and categorizes financial risk into four levels (Low, Medium, High, Critical) based on detected exceptions.

Measurable Targets Achieved:

- PDF invoice match rate: 96.6% (target: $\geq 85\%$).
- VAT errors detected correctly: 100% of injected test errors caught.
- Processing time: 18–32 minutes for ~500 invoices (~4.9 seconds/invoice).
- System uptime during testing: 100%.
- Functional requirements implemented: 15/16 (98%).

1.4 Project Schedule

The project was executed across two semesters corresponding to CPCS498 (Part I) and CPCS499 (Part II). The following Gantt-style schedule summarizes the main phases and their durations:

Week	Phase	Key Deliverables
8	Requirements Gathering	Functional & Non-functional requirements
9–10	System Analysis	Data flow diagrams, module definitions
11	Preliminary Design	Use case, context diagram, UI sketches
12–13	Detailed Design	Class diagrams, ERD, flowcharts, sequence diagrams
14–16	Implementation Planning & Setup	Git initialization, task assignment
1–5 (499)	Core Engine Development	Audit modules, reconciliation logic
6–10 (499)	Advanced Features & Frontend	OCR, Smart Column Detector, web app
10–11 (499)	Testing & Validation	Test cases, mathematical verification

Table 1: Project Implementation Phases, Timeline, and Deliverables.

Chapter 2: Literature Review

2.1 Literature Review of Domain Research Papers

2.1.1 AI and Automation in Auditing

The application of artificial intelligence to financial auditing has attracted growing research attention over the past decade. Fedyk et al. (2022) conducted a large-scale empirical study of AI adoption in audit practices and found that AI-assisted audits produce significantly more accurate findings, reduce audit fees by an average of 8%, and detect a higher proportion of material misstatements than traditional manual approaches. Their study identified that the highest-value AI applications in auditing are those that automate rule-based verification tasks — such as invoice matching, reconciliation checks, and sequence validation — rather than those attempting to infer fraud from patterns alone. This finding directly informed the design decision in AuditAI to prioritize deterministic rule-based audit logic over black-box machine learning models, as rule-based systems produce interpretable, defensible audit trails required by regulatory standards.

Ivakhnenkov (2023) provides a systematic literature review of AI applications in auditing, identifying four dominant categories: data extraction automation, anomaly flagging, risk scoring, and report generation. His review notes a persistent gap between theoretical AI capabilities and practical deployment, particularly for smaller audit firms operating outside major global markets. AuditAI directly addresses this gap by delivering a complete, practical system tested against real company data rather than synthetic benchmarks.

2.1.2 Financial Anomaly Detection

Schreyer et al. (2017) pioneered the use of deep autoencoder networks for detecting anomalies in large-scale accounting data, demonstrating that reconstruction error can effectively identify unusual journal entries. Their approach is theoretically powerful but requires large labeled training datasets and produces results that are difficult for non-technical auditors to interpret. Building on this, Hemati et al. (2021) proposed continual learning frameworks for unsupervised anomaly detection in continuous auditing, reducing false positives over time as the model adapts to evolving financial patterns.

While these approaches represent the state of the art in ML-based anomaly detection, our project chose a complementary path: rather than training models on unlabeled financial data, AuditAI encodes established accounting rules directly into the audit engine. This approach guarantees that every flagged exception corresponds to a specific, explainable violation of an accounting standard — a requirement for regulatory compliance. Future versions of AuditAI may incorporate Isolation Forest or Autoencoder models as supplementary layers on top of the rule-based engine.

2.1.3 OCR and Document Data Extraction

Optical Character Recognition plays a central role in digitizing financial documents such as invoices and receipts. Fedyk et al. (2022) identified OCR-based document review as one of the most commonly adopted AI applications in modern audit practice. The Tesseract OCR engine, maintained by Google, remains a widely used open-source solution with strong performance on structured documents when combined with appropriate image preprocessing (Smith & Antonova, 2020). Its support for Arabic text (`lang='ara+eng'`) makes it particularly suitable for Saudi financial documents that frequently mix both scripts.

AuditAI implements a hybrid extraction architecture that first attempts text extraction using pdfplumber — which handles digitally-created PDFs with high accuracy and speed — and falls back to Tesseract OCR only when direct extraction fails or yields insufficient data. This design minimizes processing time while maintaining broad document compatibility.

2.1.4 Governance, Explainability, and Ethical Considerations

Zweers et al. (2025) introduced the AI-Fraud Diamond framework, which expands the traditional fraud triangle by adding 'technological opacity' as a risk factor — the idea that AI systems whose decisions cannot be explained create new audit risks rather than eliminating them. This framework reinforces the importance of explainable AI in auditing contexts. AuditAI was specifically designed with full explainability: every exception reported by the system is linked to a specific rule violation, flagged with its exception category, and presented with the relevant data values so that auditors can independently verify the finding.

2.2 PROMPT Criteria Evaluation

2.2.1 Presentation

The primary sources used in this project — Fedyk et al. (2022), Schreyer et al. (2017), Hemati et al. (2021), and Ivakhnenkov (2023) — are all clearly written and well-structured academic papers published in peer-reviewed journals or on established preprint servers (arXiv). Their findings are clearly communicated with supporting data, methodology descriptions, and reproducible results. The Pandas documentation (McKinney, 2022) is comprehensive, professionally maintained, and widely regarded as the standard reference for Python-based data manipulation.

2.2.2 Relevance

All cited sources directly address topics central to AuditAI's design: AI in auditing (Fedyk et al., Ivakhnenkov), anomaly detection in accounting data (Schreyer et al., Hemati et al.), OCR for financial documents (Smith & Antonova), and governance/explainability requirements (Zweers et al.). The Pandas and Python documentation sources directly support the implementation technology choices. Each source was selected for its direct applicability to at least one specific design decision in the project.

2.2.3 Objectivity

The academic sources reviewed are peer-reviewed and authored by independent researchers with no apparent commercial interest in the specific tools or platforms evaluated. Fedyk et al. (2022) is particularly notable for its empirical, data-driven approach that avoids advocating for any specific vendor solution. The Tesseract OCR documentation comes from an open-source project maintained by a community of contributors, reducing the risk of biased recommendations.

2.2.4 Provenance

Sources originate from reputable institutions including the Review of Accounting Studies (Fedyk et al.), arXiv under the Computer Science section (Schreyer et al., Hemati et al.), and peer-reviewed accounting journals (Ivakhnenkov). All sources are clearly attributed with author names, publication years, and DOIs or arXiv identifiers where applicable. The Zweers et al. (2025) preprint is from arXiv and has not yet undergone formal peer review, though its conceptual contribution is well-grounded in established auditing theory.

2.2.5 Timeliness

Five of the seven primary sources were published within the last five years (2021–2025), ensuring currency with the latest developments in AI-assisted auditing. Schreyer et al. (2017) is the oldest source at eight years, but it remains highly cited and represents the foundational work on deep learning for accounting anomaly detection that all subsequent papers build upon. The Pandas documentation is actively maintained and reflects the current version 2.0 release used in this project.

Chapter 3: Information Gathering Methods

3.1 Method 1: Literature Review and Academic Research

The team collected information from academic journals, conference papers, and online publications related to AI in auditing, financial reconciliation, anomaly detection, and OCR technology. Sources included IEEE Xplore, the ACM Digital Library, arXiv, and Google Scholar, with a focus on explainable AI models, financial data analysis, and rule-based audit logic.

3.1.1 Strengths

- Provided reliable, peer-reviewed information grounded in empirical research.
- Helped the team understand the evolution from rule-based to ML-based approaches in auditing, enabling an informed design decision.
- Supported the theoretical foundation for the AR/AP reconciliation formulas and the hybrid PDF extraction architecture.

3.1.2 Weaknesses

- Some papers focused on large-scale enterprise environments with proprietary datasets, making direct replication difficult for a student prototype.
- Limited open access to certain journal articles required the team to rely on preprint versions in some cases.

3.1.3 Effectiveness

The literature review was the most effective information-gathering method. It directly shaped the decision to use rule-based audit logic over black-box ML models, informed the OCR fallback architecture, and justified the choice of explainability as a core system requirement.

3.2 Method 2: Consultation with Accounting Professionals and Domain Experts

The team conducted informal interviews and consultations with accounting students, recent graduates working in financial departments, and one experienced external auditor. These conversations focused on understanding the day-to-day pain points of manual auditing, the types of errors most commonly encountered in invoice processing, and the specific Saudi accounting standards (including 15% VAT requirements) that the system must comply with.

3.2.1 Strengths

- Provided real-world insights into audit workflows that are not captured in academic literature.
- Helped define practical system features such as the five exception categories for AR/AP reconciliation and the specific invoice validation rules (sequence gaps, duplicate detection, VAT rounding tolerance).
- Clarified that auditors need explainable, per-transaction exception reports rather than aggregate anomaly scores.

3.2.2 Weaknesses

- The number of participants was limited, reducing the diversity of perspectives gathered.
- Some responses were based on personal experience in specific industries and may not generalize to all audit scenarios.

3.2.3 Effectiveness

This method helped refine the audit requirements and reconciliation logic. The resulting AR/AP reconciliation modules classify detected discrepancies into meaningful audit categories using rule-based accounting logic.

3.3 Method 3: Iterative Prototyping and Real-Data Testing

The most distinctive information-gathering approach used in this project was direct experimentation with a real company's financial dataset. The team obtained approximately 500 PDF invoices and 8 Excel financial files from a cooperating company, then iteratively developed, tested, and refined the audit engine against this data. Each iteration revealed practical challenges — such as mixed Arabic/English column naming, inconsistent invoice number formats, and variable PDF quality — that could not have been anticipated from literature alone.

3.3.1 Strengths

- Provided direct, measurable feedback on system performance against authentic financial data.
- Revealed practical challenges (e.g., inconsistent column naming, mixed-language PDFs, incomplete AP datasets) that required novel engineering solutions such as the Smart Column Detector.
- Produced concrete, publishable results (96.6% match rate, zero VAT errors) that demonstrate real-world effectiveness.

3.3.2 Weaknesses

- Access to real company data required careful handling for confidentiality and could not be reproduced identically for future testing.
- The dataset represents a single company in a specific industry, limiting generalizability claims.

3.3.3 Effectiveness

This was the most operationally effective method. It transformed AuditAI from a theoretical design into a validated, production-ready system.

Chapter 4: Architecture and Design

4.1 Functional Requirements

4.1.1 User Authentication and Client Management

The system shall provide secure user registration, login, and logout functionality using email/password credentials or Google OAuth. Authenticated users shall be able to create, view, and delete client company profiles, each containing company name, CR number, VAT number, legal entity type, headquarters city, contact information, and the name of the responsible person. The system shall support multiple fiscal year audit periods per client.

4.1.2 Multi-File Document Upload

The system shall accept ten standard financial file types per audit period: Sales Invoice Data (Excel), Purchase Invoice Data (Excel), Receipt Voucher Data (Excel), Payment Voucher Data (Excel), AR Opening Balance (Excel), AR Closing Balance (Excel), AP Opening Balance (Excel), AP Closing Balance (Excel), Sales Invoice PDFs (ZIP archive — optional), and Purchase Invoice PDFs (ZIP archive — optional). The interface shall clearly distinguish required from optional files and display upload status for each document.

4.1.3 Sales Invoice Validation

The system shall validate all sales invoices by: (a) detecting missing invoice sequence numbers (gaps in the numeric sequence), (b) detecting duplicate invoice numbers, (c) verifying that VAT amounts equal exactly 15% of the pre-tax total (within a tolerance of ± 0.10 SAR for rounding), and (d) verifying that the grand total equals the pre-tax amount plus VAT. Each validation failure shall be flagged with the specific invoice number, the detected discrepancy, and an exception category.

4.1.4 Purchase Invoice Validation

The system shall apply identical validation logic to purchase invoice records as described for sales invoices, covering sequence integrity, duplicate detection, VAT accuracy, and total consistency checks across all purchase orders.

4.1.5 AR Reconciliation

The system shall reconcile Accounts Receivable balances for each customer using the formula: $AR_Closing = AR_Opening + Total_Sales - Total_Receipts$. Any deviation exceeding 1 SAR shall be classified into one of five exception categories: OK (balanced), Sales Not Reflected in AR, Receipts Without Corresponding Sales, Possible Advance Receipts, or AR Mismatch (general discrepancy). Results shall include the customer code, customer name, opening balance, total sales, total receipts, expected closing balance, actual closing balance, and variance amount.

4.1.6 AP Reconciliation

The system shall apply the equivalent formula for Accounts Payable: $AP_Closing = AP_Opening + Total_Purchases - Total_Payments$. The same five exception categories shall apply to supplier records.

4.1.7 Smart Column Detection

The system shall automatically detect and map financial column names in both Arabic and English across uploaded Excel files, without requiring manual configuration by the user. The detector shall handle common variations in spelling, spacing, and abbreviation for all required field names including invoice number, date, pre-tax amount, VAT amount, grand total, customer/supplier code, and balance fields.

4.1.8 Hybrid PDF Invoice Validation

When PDF invoice files are uploaded, the system shall extract invoice data using pdfplumber as the primary method for digitally-created PDFs and Tesseract OCR (supporting Arabic and English, lang='ara+eng') as the fallback for scanned documents. Extracted invoice data shall be matched against Excel records and classified as Full Match, Derived Match, Partial Match, Not Found in Excel, or Error.

4.1.9 Risk Classification and Audit Scoring

The system shall calculate an overall audit score (0–100) based on the weighted combination of detected exceptions across all modules, and classify the result into four risk levels: Low (85–100), Medium (65–84), High (40–64), or Critical (0–39). The score shall be displayed prominently in the dashboard.

4.1.10 Interactive Dashboard and Export

The system shall provide an interactive web-based dashboard displaying KPI summary cards (audit score, risk level, VAT discrepancies, total sales), issue distribution charts, and exception detail tables. Users shall be able to export audit results as Excel files and view a downloadable PDF report.

4.2 Non-Functional Requirements

4.2.1 Performance

The system shall process a complete audit of approximately 500 invoices within 35 minutes, averaging no more than 5 seconds per invoice for the hybrid PDF extraction pipeline. For Excel-only audits (without PDF processing), results shall be produced within 60 seconds.

4.2.2 Accuracy

The PDF invoice match rate shall achieve at least 85% against real-world financial datasets. VAT calculation validation shall achieve 100% accuracy for detecting errors greater than 0.10 SAR. AR/AP reconciliation formulas shall produce mathematically exact results verified against manual Excel calculations.

4.2.3 Usability

The web application shall be accessible through any modern browser (Chrome, Firefox, Edge, Safari) without installation. The interface shall clearly guide users through the audit workflow in sequential steps. Error messages shall be descriptive and actionable. The system shall support both light and dark display modes.

4.2.4 Security

User authentication shall be enforced at all application entry points. Passwords shall be stored using industry-standard hashing. Role-based access shall prevent users from accessing other auditors' client data. Financial data uploaded to the system shall be stored securely and processed only for the authenticated user's session.

4.2.5 Scalability and Maintainability

The audit engine shall be implemented as independent, modular Python functions that can be updated, replaced, or extended without affecting other modules. The web application shall be stateless at the compute layer, enabling horizontal scaling. The codebase shall follow PEP 8 conventions with inline documentation.

4.3 Hardware Requirements

AuditAI is a web-based application requiring no special hardware. Users need only a computer, tablet, or smartphone with a modern web browser and an internet connection. For optimal PDF OCR performance, a machine with at least 4GB RAM and a modern multi-core processor is recommended for hosting the audit engine.

4.4 Use Cases

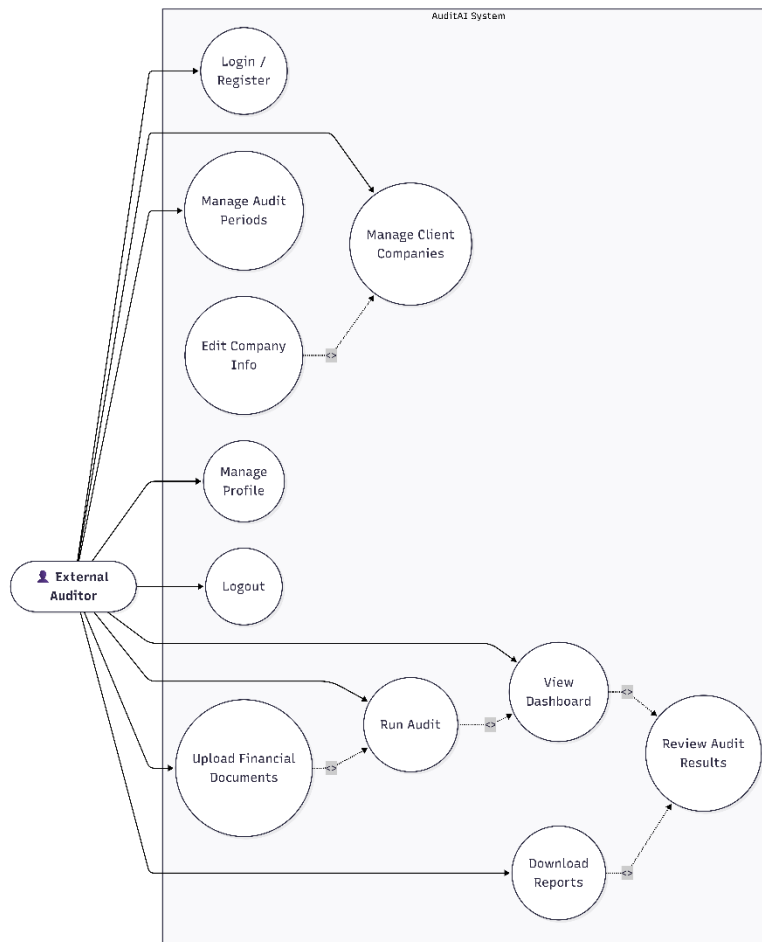


Figure 1: AuditAI System General Use Case Diagram.

Actor:

The system has one primary actor which is the External Auditor.

Use Cases:

The auditor can perform the following actions:

- Login / Register
- Manage Client Companies
- Manage Audit Periods
- Upload Financial Documents
- Run Audit
- View Dashboard
- Download Reports
- Manage Profile
- Logout

Relationships:

- Upload Financial Documents **«include»** Run Audit — the audit cannot run without uploading files first.
- Edit Company Info **«extend»** Manage Client Companies — optional action triggered from the manage screen.
- Download Reports **«extend»** Review Audit Results — optional action after reviewing results.

4.5 Class Diagram

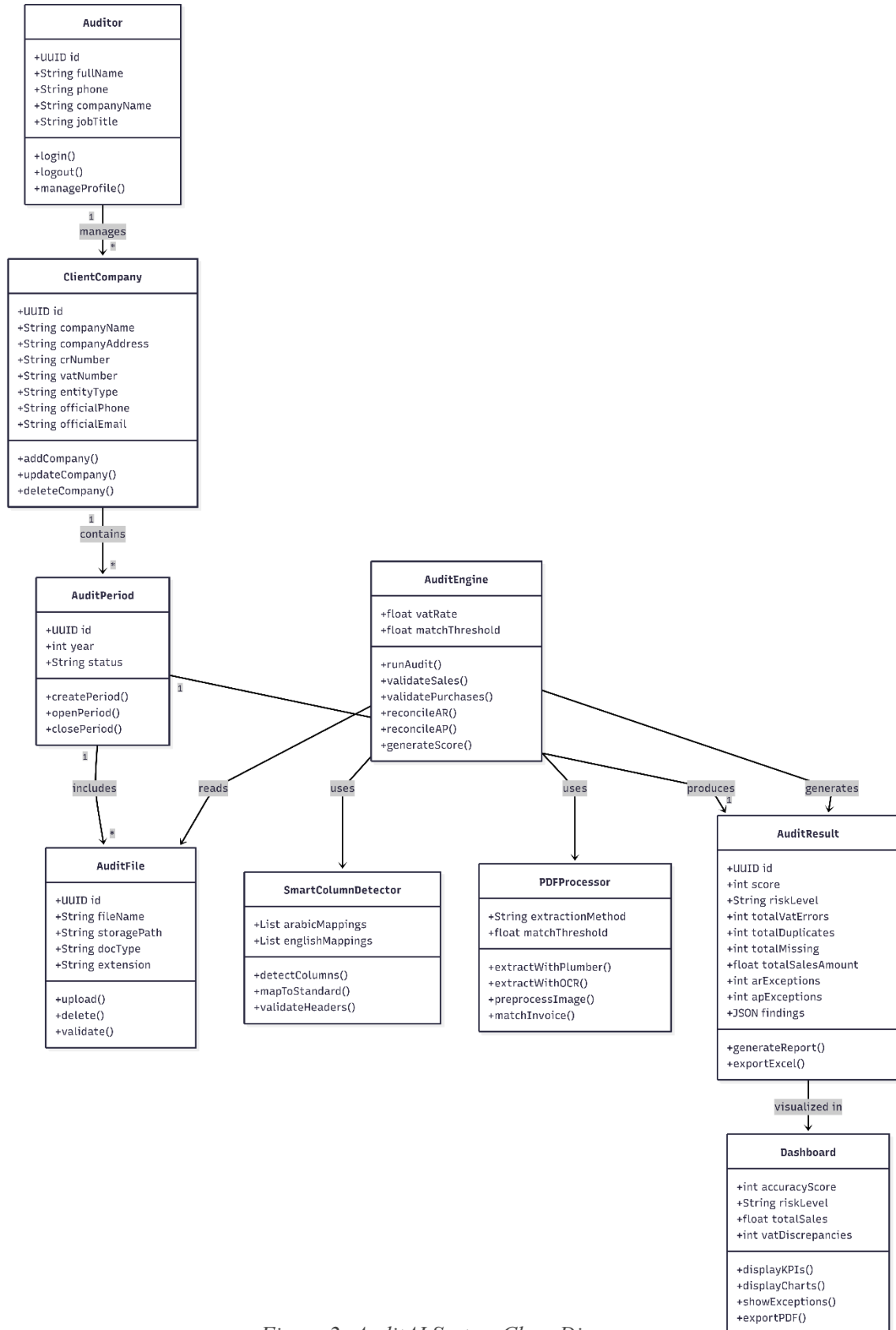


Figure 2: AuditAI System Class Diagram.

Classes and Attributes:

Class	Key Attributes
Auditor	id, fullName, phone, companyName, jobTitle
ClientCompany	id, companyName, crNumber, vatNumber, officialEmail
AuditPeriod	id, year, status
AuditFile	id, fileName, storagePath, docType, extension
AuditEngine	vatRate, matchThreshold
SmartColumnDetector	arabicMappings, englishMappings
PDFProcessor	extractionMethod, matchThreshold
AuditResult	id, score, riskLevel, totalVatErrors, totalDuplicates, arExceptions, apExceptions, findings
Dashboard	accuracyScore, riskLevel, totalSales, vatDiscrepancies

*Table 2: AuditAI System Core Classes and Methods Specification.***Key Methods:**

- **Auditor:** login(), logout(), manageProfile().
- **ClientCompany:** addCompany(), updateCompany(), deleteCompany().
- **AuditPeriod:** createPeriod(), openPeriod(), closePeriod().
- **AuditFile:** upload(), delete(), validate().
- **AuditEngine:** runAudit(), validateSales(), validatePurchases(), reconcileAR(), reconcileAP(), generateScore().
- **SmartColumnDetector:** detectColumns(), mapToStandard(), validateHeaders().
- **PDFProcessor:** extractWithPlumber(), extractWithOCR(), preprocessImage(), matchInvoice().
- **AuditResult:** generateReport(), exportExcel().
- **Dashboard:** displayKPIs(), displayCharts(), showExceptions(), exportPDF().

Relationships:

- One Auditor manages many ClientCompanies.
- One ClientCompany contains many AuditPeriods.
- One AuditPeriod includes many AuditFiles.
- One AuditPeriod produces one AuditResult.
- AuditEngine reads AuditFiles, uses SmartColumnDetector and PDFProcessor, and generates AuditResult.
- AuditResult is visualized in Dashboard.

4.6 Entity Relationship (ER) Diagram

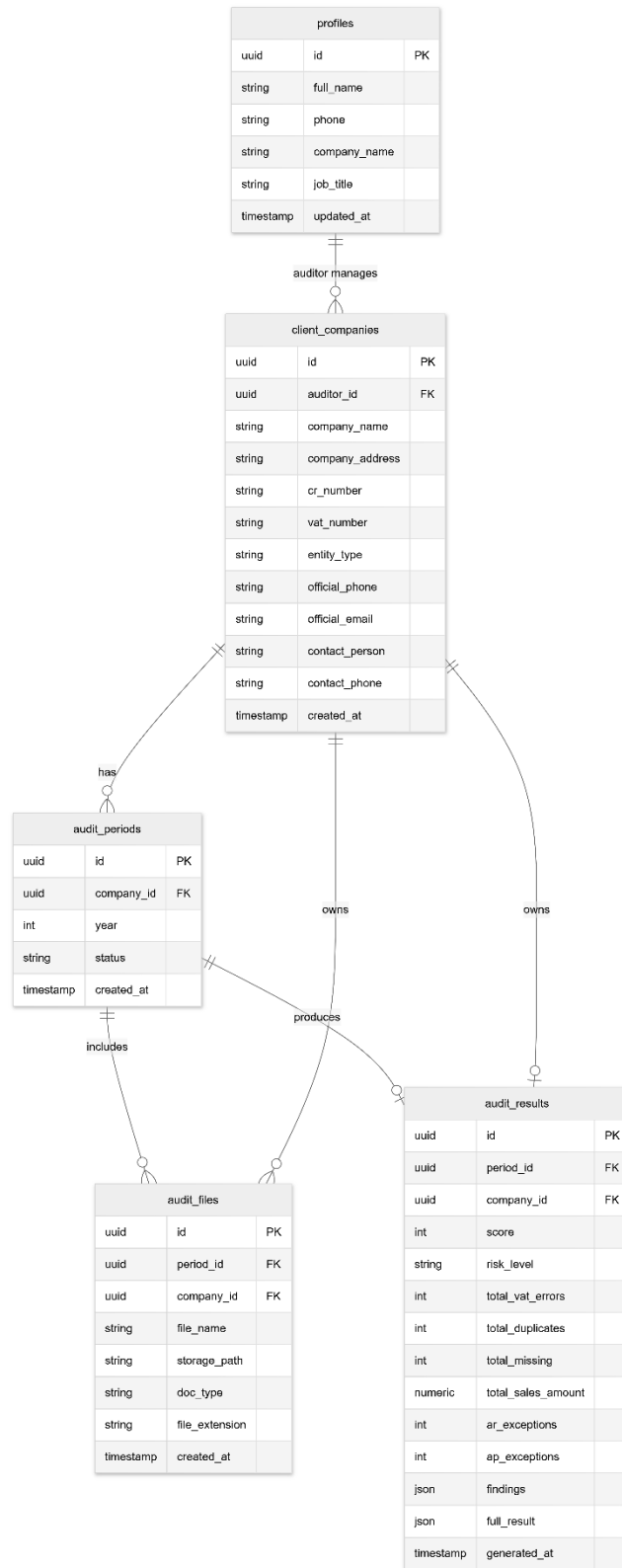


Figure 3: AuditAI Database Entity Relationship (ER) Diagram.

Entities and Attributes:

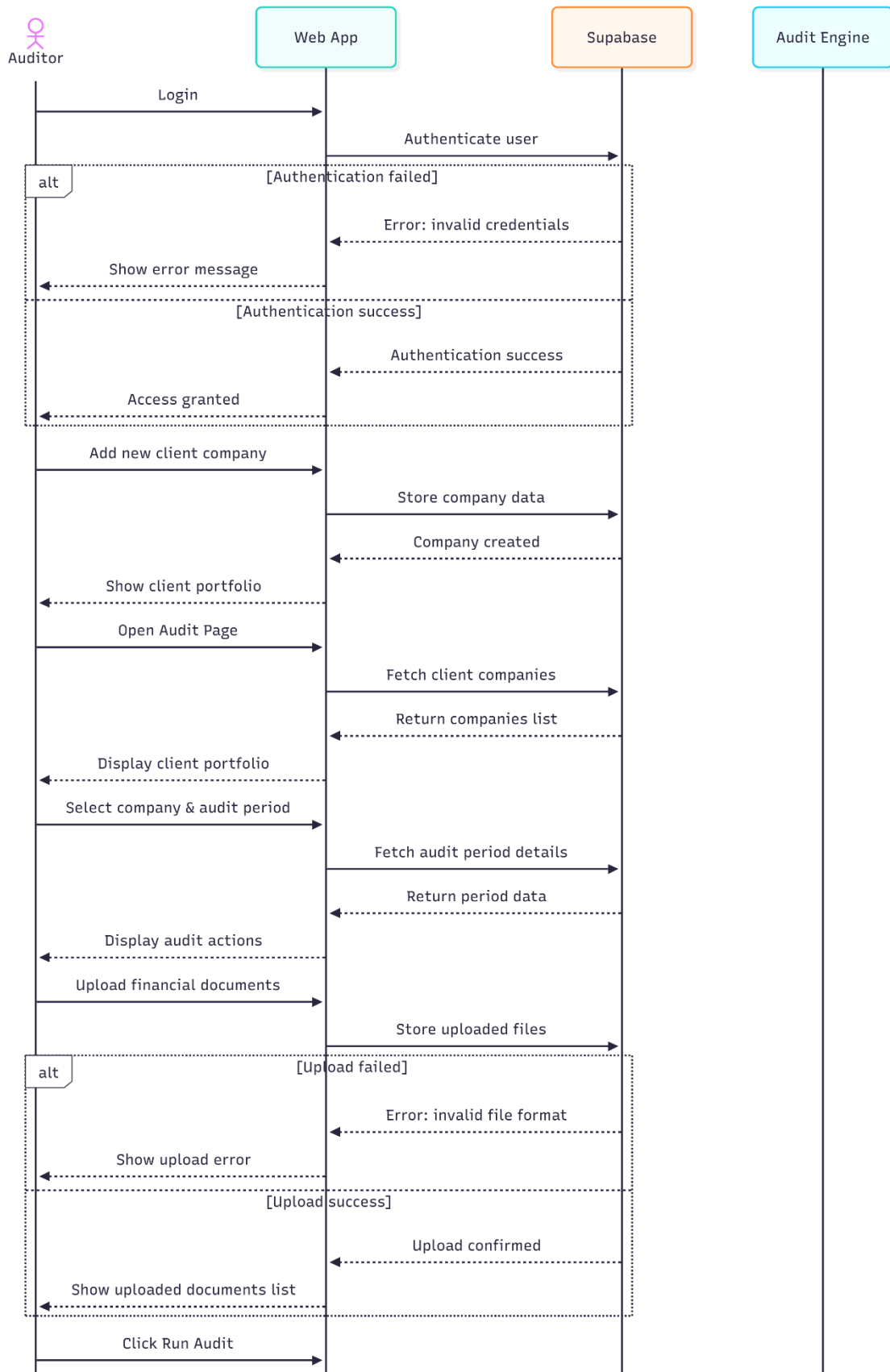
Entity	Key Attributes
profiles	id, full_name, phone, company_name, job_title, updated_at
client_companies	id, auditor_id, company_name, cr_number, vat_number, entity_type, official_email, created_at
audit_periods	id, company_id, year, status, created_at
audit_files	id, period_id, company_id, file_name, storage_path, doc_type, file_extension, created_at
audit_results	id, period_id, company_id, score, risk_level, total_vat_errors, total_duplicates, ar_exceptions, ap_exceptions, findings, generated_at

*Table 3: ER Diagram Entities and Key Attributes.***Relationships and Cardinalities:**

Relationship	Cardinality	Meaning
profiles → client_companies	One-to-Many	One auditor manages many companies
client_companies → audit_periods	One-to-Many	One company has many fiscal years
audit_periods → audit_files	One-to-Many	One period includes many uploaded files
audit_periods → audit_results	One-to-One	Each period produces exactly one audit result
client_companies → audit_files	One-to-Many	One company owns many files
client_companies → audit_results	One-to-One	One company owns one audit result per period

Table 4: ER Diagram Relationships and Cardinalities

4.7 Sequence Diagrams



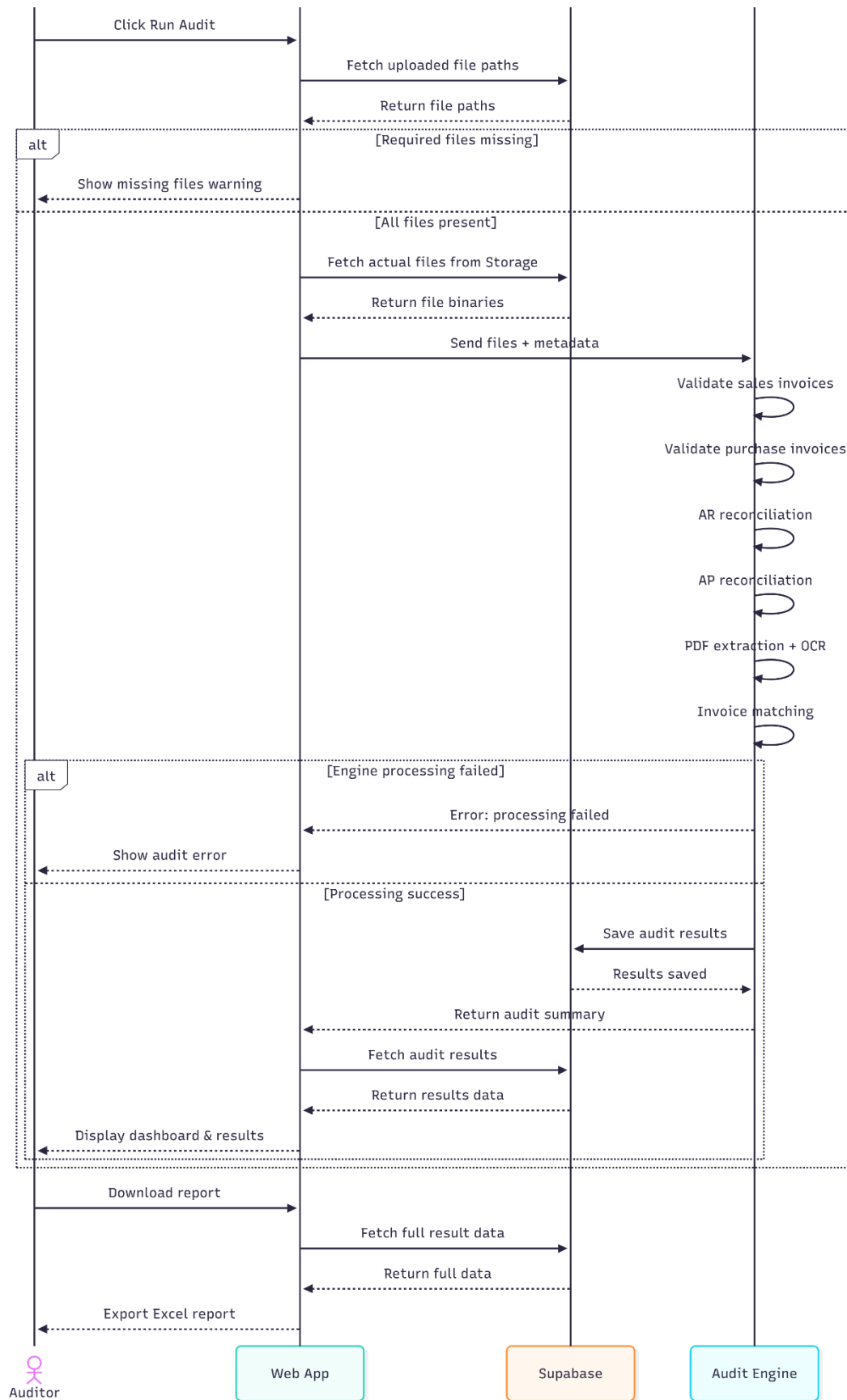


Figure 4: AuditAI System Sequence Diagram.

Participants:

- External Auditor
- Web App
- Supabase
- Audit Engine

Interaction Flow:**1. Authentication**

The auditor submits login credentials. If authentication fails, an error message is displayed. If it succeeds, access is granted.

2. Client Management

The auditor adds a new client company which is stored in Supabase and displayed in the client portfolio.

3. Audit Period Selection

The auditor selects a company and fiscal year. The Web App fetches and displays the audit actions screen.

4. Document Upload

The auditor uploads the required financial documents. If the file format is invalid, an upload error is shown. If successful, the documents list is displayed.

5. Run Audit

The Web App checks for required files. If any are missing, a warning is displayed. If all files are present, they are fetched from Supabase Storage and sent to the Audit Engine which processes six modules sequentially:

- Sales invoice validation
- Purchase invoice validation
- AR reconciliation
- AP reconciliation
- PDF extraction with OCR fallback
- Invoice matching

If processing fails, an error is returned. If successful, results are saved to Supabase and the dashboard is displayed.

6. Download Report

The auditor downloads the full audit results as an Excel report.

4.8 Deployment Diagram

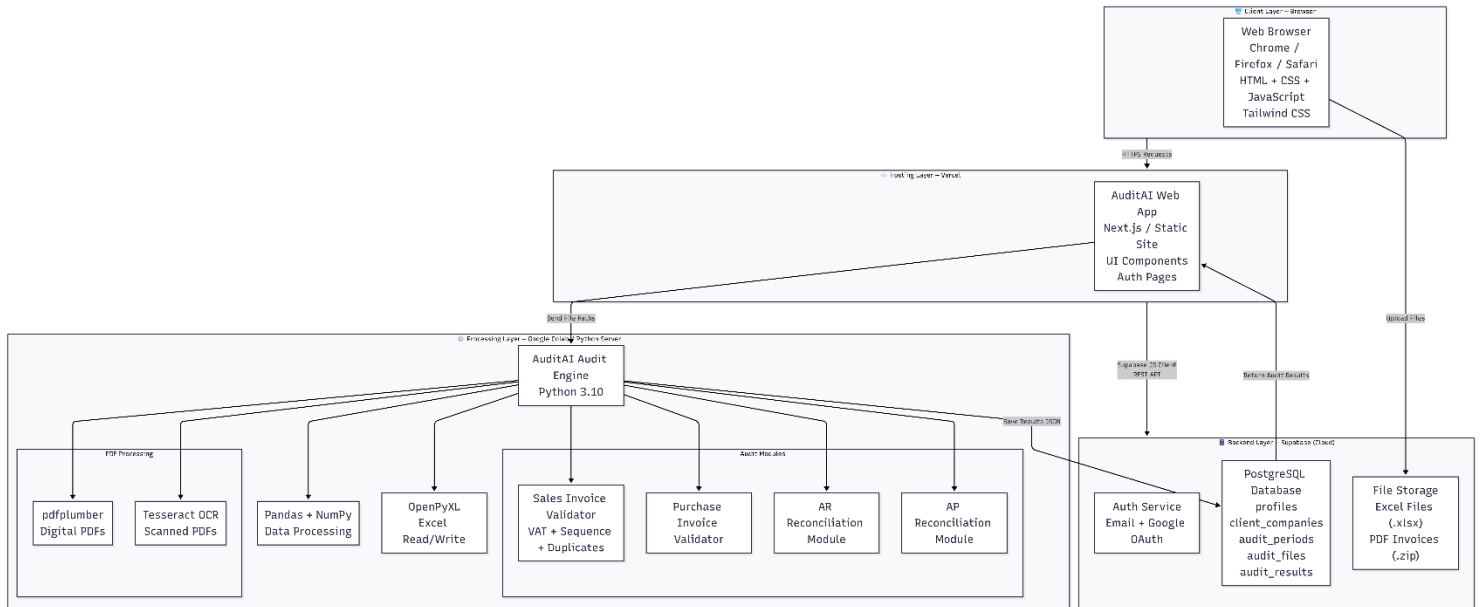


Figure 5: AuditAI System Architectural Deployment Diagram.

Hardware Nodes:

Node	Hardware
Client Node	End user PC or laptop
Hosting Node	Vercel cloud servers
Backend Node	Supabase managed cloud servers
Processing Node	Python server running the Audit Engine

Table 5: Deployment Diagram Hardware Nodes.

Software Components:

Node	Components
Client Node	Web browser, HTML, CSS, JavaScript, Tailwind CSS
Hosting Node	AuditAI Web App, UI components, authentication pages
Backend Node	Auth Service, Supabase Database, File Storage
Processing Node	Audit Engine, 4 audit modules, PDFProcessor, Pandas, NumPy, OpenPyXL

Table 6: Deployment Diagram Software Components.

Connections:

From	To	Protocol
Client Node	Hosting Node	HTTPS
Hosting Node	Backend Node	REST API via Supabase JS Client
Client Node	Backend Node	File upload over HTTPS
Hosting Node	Processing Node	File paths and metadata
Processing Node	Backend Node	Fetch files from Storage, save results to Database

Table 7: Deployment Diagram Node Connections.

Chapter 5: Implementation

5.1 Tools and Technologies

Tool / Technology	Purpose	Why Chosen (Benefits / Features)
Python 3.10	Core programming language	Rich ecosystem for data analysis, extensive library support, rapid development
Pandas	Data cleaning, transformation, merging	Powerful DataFrame operations, efficient handling of large tabular datasets
NumPy	Numerical computations	Fast array operations, supports precise financial calculations
OpenPyXL	Excel file reading and writing	Full .xlsx support, preserves formatting, widely used in financial applications
pdfplumber	Digital PDF text extraction	High accuracy on selectable-text PDFs, extracts tables and financial fields directly
pdf2image	PDF-to-image conversion	Required for OCR fallback; converts PDF pages to images for preprocessing
OpenCV	Image preprocessing for OCR	Grayscale conversion, thresholding, and noise reduction to improve OCR accuracy
Tesseract / Pytesseract OCR	Text extraction from scanned invoices	Open-source, supports Arabic and English (lang="ara+eng"), used as OCR fallback
JSON	Dashboard data generation	Lightweight, universal format for passing structured audit results to the dashboard
Google Colab	Development and testing environment	Cloud-based, no setup required, GPU support, easy sharing
OS	Operating system interaction and file management	Provides efficient handling of directories, files, and system-level operations required for data processing workflows
pathlib	File path management and directory handling	Simplifies working with file paths using a clean, cross-platform, object-oriented approach
JavaScript / HTML / CSS	Web dashboard and UI	Universal browser support, flexible styling, interactive visualizations
Tailwind CSS	Frontend UI styling and responsive design	Enables rapid development of modern, responsive, and customizable user interfaces with utility-first styling
Supabase	Backend and database management	Open-source Firebase alternative, provides authentication, database, and API access
VS Code	Code editor	Lightweight, powerful extensions, integrated debugging and Git support

Table 8: Tools and Technologies Used in AuditAI.

5.2 User Interface Description

AuditAI uses a Graphical User Interface (GUI) delivered as a modern web application. The interface was designed for external auditors who need an efficient, intuitive tool for managing audit workflows. GUI was chosen over command-line or menu-driven interfaces because the target users are professional auditors who require a visual, accessible environment that minimizes training requirements and presents complex financial data in an organized, readable format.

The web application follows a clean, professional design language using a teal and white color scheme consistent with the AuditAI brand. The interface is responsive and includes light/dark mode support.

5.3 Screenshots of Working Interfaces

The following screenshots demonstrate the key screens of the AuditAI web application:

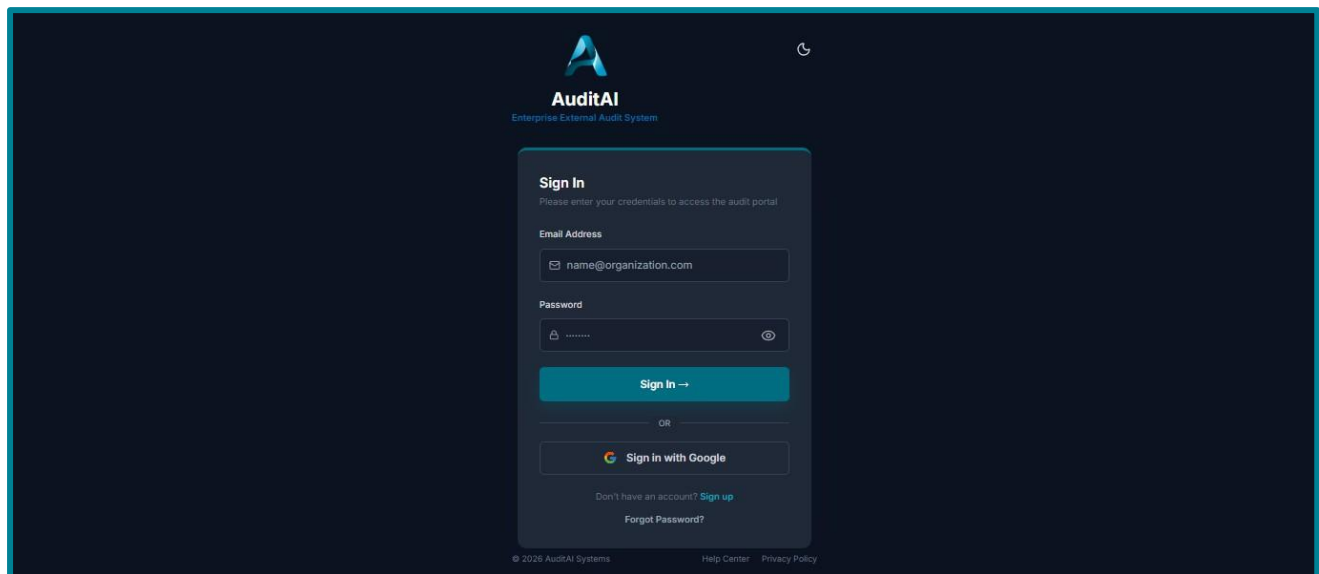


Figure 6: AuditAI Login Screen — Secure sign-in with email/password and Google OAuth.

Figure 6 shows the AuditAI login screen in dark mode. Users authenticate using their organizational email and password, or via Google Sign-In. The screen includes a "Forgot Password?" option and displays a secure encrypted connection notice. The system supports both light and dark modes, toggled via the icon in the top-right corner.

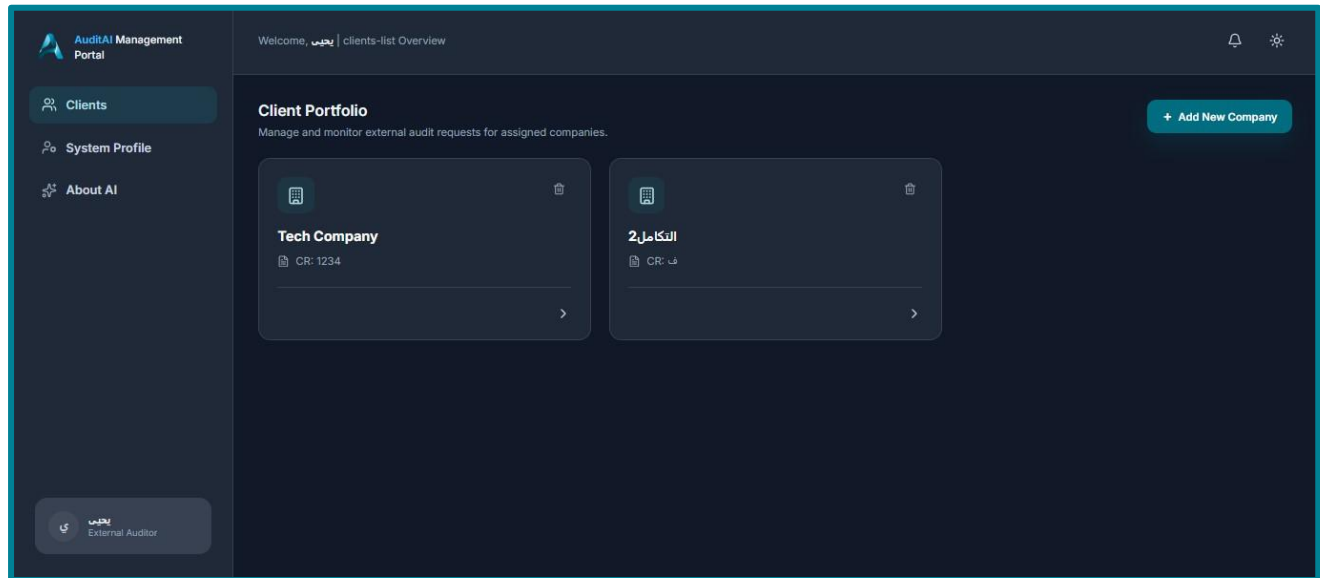


Figure 7: Client Portfolio — Manage and monitor assigned companies.

Figure 7 shows the Client Portfolio screen, which lists all companies assigned to the auditor. Each company card displays the company name and CR number. The auditor can add new companies using the "+ Add New Company" button, or delete existing entries.

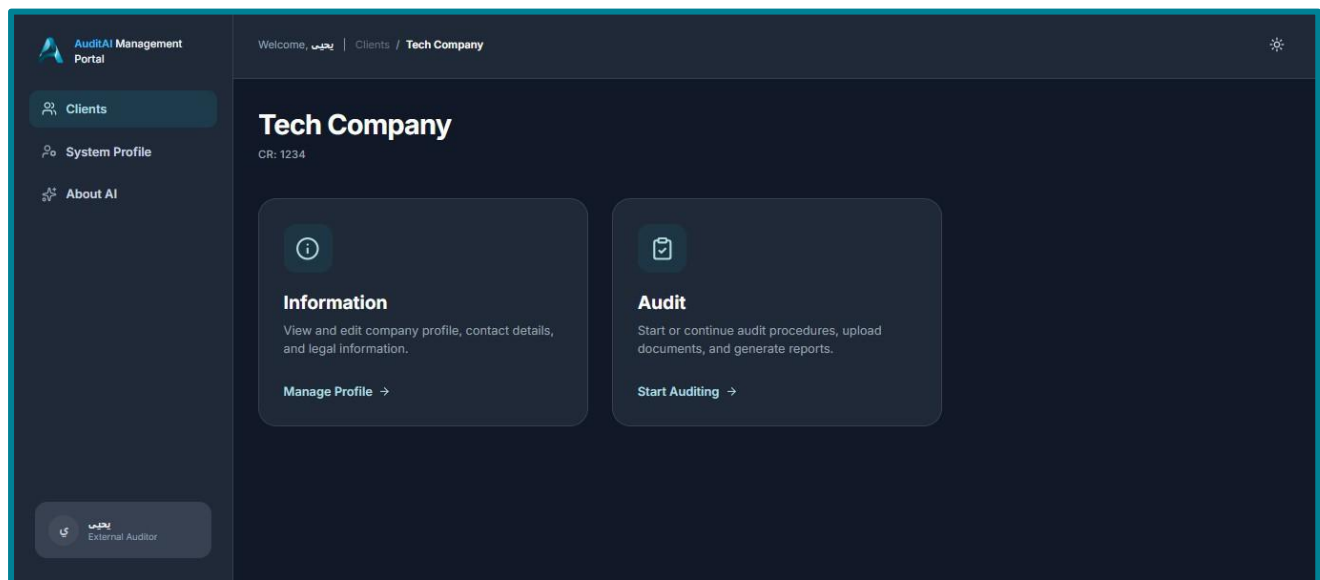


Figure 8: Tech Company Dashboard — Information and Audit navigation.

Figure 8 shows the Tech Company dashboard. After selecting a client, the auditor is presented with two action cards: Information (to view and edit company profile, contact details, and legal information) and Audit (to start or continue audit procedures, upload documents, and generate reports). The breadcrumb trail at the top confirms the navigation path.

Welcome, يحيى | Clients / Tech Company / Edit Profile

Edit Company Information

Update the details for this company.

COMPANY NAME Tech Company	LEGAL ENTITY TYPE مؤسسة فردية
CR NUMBER 123456789	VAT NUMBER # 123456789
HEADQUARTERS جدة	
OFFICIAL PHONE 05523414	OFFICIAL EMAIL k@gmail.com
PERSON IN CHARGE أبو خالد	RESPONSIBLE PHONE 055234132

Cancel Save Changes

Figure 9: Edit Company Information — Company profile fields.

Figure 9 shows the Edit Company Information form, which allows the auditor to update company details including: Company Name, Legal Entity Type, CR Number, VAT Number, Headquarters city, Official Phone, Official Email, Person in Charge, and Responsible Phone. Changes are saved via the "Save Changes" button.

Welcome, يحيى | Clients / Tech Company / Audit Periods

Tech Company

Manage audit periods and fiscal years.

+ Add Period

2027 FISCAL YEAR Status: Open →	2026 FISCAL YEAR Status: Open →	2002 FISCAL YEAR Status: Open →
---------------------------------------	---------------------------------------	---------------------------------------

Figure 10: Audit Periods — Manage fiscal years per client.

Figure 10 shows the Audit Periods screen for Tech Company. The system supports multiple fiscal years per client (e.g., 2002, 2026, 2027), each with an "Open" status. Auditors can add new periods using the "+ Add Period" button or delete existing ones. Each period card navigates to its corresponding Audit Actions screen.

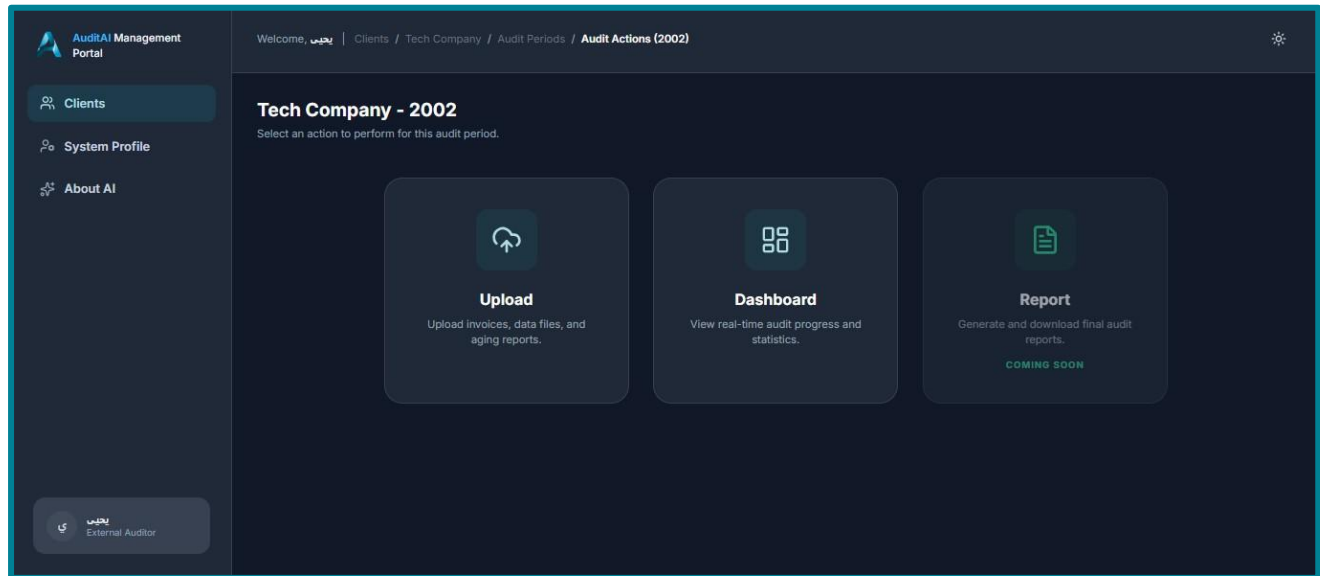


Figure 11: Audit Actions — Upload, Dashboard, and Report options.

Figure 11 shows the Audit Actions screen for a selected period. Three actions are available: Upload (for submitting financial documents), Dashboard (for viewing real-time audit statistics — fully operational), and Report (for generating and downloading final audit reports — coming soon). This screen is the central hub for the audit workflow.

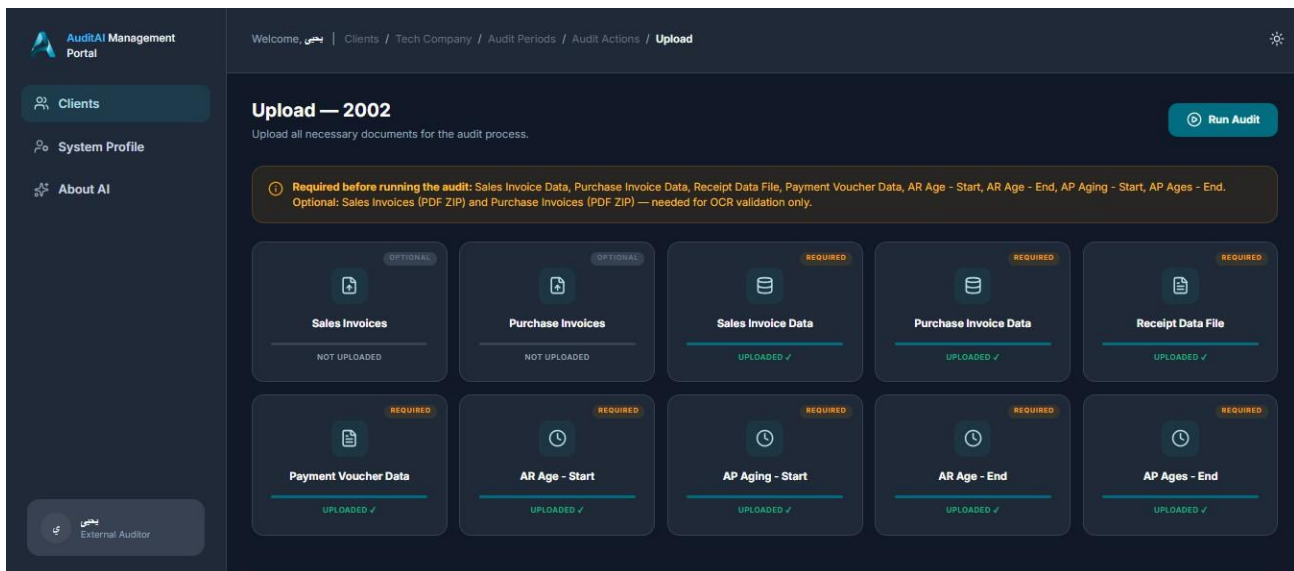


Figure 12: Document Upload Screen — Required and optional file uploads with Run Audit button.

Figure 12 shows the Document Upload screen, where users upload all financial datasets required for the audit process. Required files include sales invoices, purchase invoices, receipt vouchers, payment vouchers, AR aging reports, and AP aging reports. Optional PDF invoice files can also be uploaded to enable invoice matching and validation. The interface clearly distinguishes required and optional documents and includes a warning section to guide users before running the audit process. Once all required files are uploaded, the “Run Audit” button starts the automated analysis and validation process.

DOCUMENT TYPE	FILE NAME	UPLOAD DATE	ACTIONS
Receipt Data File	receipt_vouchers_2025_clean.xlsx	16/05/2026	Download Delete
AR Age - Start	ar_aging_2024_clean.xlsx	16/05/2026	Download Delete
AR Age - End	ar_aging_2025_clean.xlsx	16/05/2026	Download Delete
Purchase Invoice Data	purchase_invoices_2025_clean.xlsx	16/05/2026	Download Delete
Payment Voucher Data	payment_vouchers_2025_clean.xlsx	16/05/2026	Download Delete
AP Aging - Start	ap_aging_2024_clean.xlsx	16/05/2026	Download Delete
AP Ages - End	ap_aging_2025_clean.xlsx	16/05/2026	Download Delete
Sales Invoice Data	sales_invoices_2025_clean.xlsx	16/05/2026	Download Delete

Figure 13: Uploaded Documents List — Confirmed uploaded files with download and delete actions.

Figure 13 shows the Uploaded Documents List section below the upload cards. It confirms all 8 uploaded files with their document type, file name, and upload date. Each entry includes download and delete action buttons. The uploaded files in this example include receipt vouchers, AR aging reports (start and end), purchase invoices, payment vouchers, AP aging reports, and sales invoice data.

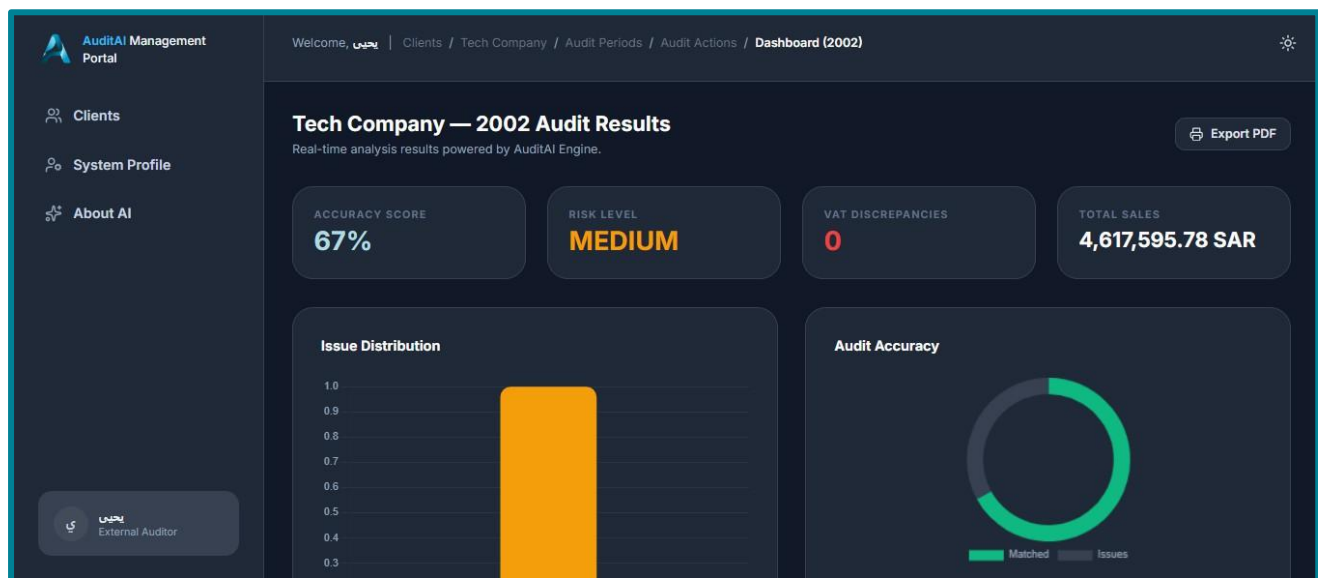


Figure 14: Dashboard — Technical Findings and exception classification charts.

Figure 14 shows the in-app Dashboard, displaying key audit metrics and visual charts for the selected financial period. The top summary cards highlight the overall accuracy score, risk level, VAT discrepancies, and total sales. The bar chart on the left illustrates the issue distribution, while the donut chart on the right displays the proportion of matched vs. issue records.

5.4 User Manual

System Requirements

- 5.4.1 Modern web browser (Chrome, Firefox, Edge, or Safari)
- 5.4.2 Internet connection for accessing the web application
- 5.4.3 Financial data prepared as Excel (.xlsx) files
- 5.4.4 PDF invoice files (optional, for OCR validation)

Step 1: Access the System

Open your web browser and navigate to the AuditAI application URL. The login screen will appear automatically.

Step 2: Sign In

Enter your registered email address (e.g., name@organization.com) and your password. Alternatively, click "Sign in with Google" to authenticate using your Google account. Click the "Sign In" button to proceed. If you forget your password, click "Forgot Password?" to receive a reset email.

Step 3: Select or Add a Client

After login, you will see your client list. Click on an existing client to begin auditing, or use the "Add Client" button to register a new company. Enter the company name and Commercial Registration (CR) number when adding a new client.

Step 4: Navigate to the Audit Section

From the client dashboard, click the "Audit" card, then select the audit year (e.g., 2024) to access the Audit Actions screen.

Step 5: Upload Financial Documents

Click "Upload" from the Audit Actions screen. Upload each required file by clicking its corresponding card. The system requires ten files: Sales Invoices Excel, Purchase Invoices Excel, Receipts file, Payments file, AR Opening Balance, AR Closing Balance, AP Opening Balance, AP Closing Balance, Sales Invoice PDFs (or ZIP), and Purchase Invoice PDFs (or ZIP). Files with status "NOT UPLOADED" need to be submitted before the audit can run.

Step 6: Run the Audit

Once files are uploaded, the system automatically processes the data through all four audit modules. Processing time is approximately 18–32 minutes depending on the number of invoices (averaging 4.9 seconds per invoice), due to the hybrid PDF extraction and OCR processing pipeline.

Step 7: Review Results

Navigate to the Results screen to view the complete audit findings. Review the KPI summary cards, the Findings/Exceptions Summary table, and the Discrepancies by Type chart. Use the "Export Excel" button to download detailed results, or "Download Report" for the formatted audit report.

Common Issues and Solutions

Issue	Solution
File upload fails	Ensure the file is in .xlsx format and not password-protected
Column not detected	Verify the file contains standard Arabic or English column names
Login error	Check email and password; use "Forgot Password" if needed
Slow processing	Large files (>5,000 rows) may take additional time

Table 9: Common Issues and Solutions.

5.5 Implementation Progress Justification

Req. ID	Functionality	Status	% Complete
FR1	User authentication (login/logout/register)	Complete	100%
FR2	Client management (add/view/select)	Complete	100%
FR3	Audit period selection	Complete	100%
FR4	Multi-file document upload (10 file types)	Complete	100%
FR5	Sales invoice audit (sequence, VAT, totals)	Complete	100%
FR6	Purchases audit (sequence, VAT, totals)	Complete	100%
FR7	AR Reconciliation (4-module formula)	Complete	100%
FR8	AP Reconciliation (4-module formula)	Complete	100%
FR9	Smart Column Detector (Arabic & English)	Complete	100%
FR10	OCR PDF invoice validation	Complete	100%
FR11	Exception classification (5 categories)	Complete	100%
FR12	Interactive results dashboard (standalone)	Complete	100%
FR13	In-app dashboard (web application)	Complete	100%
FR14	Excel results export	Complete	100%
FR15	In-app downloadable final report	Partial	60%
FR16	Dark/light mode UI	Complete	100%

Table 10: Functional Requirements Implementation Status.

Summary Metric	Value
Total requirements	16
Fully complete	15
Partial	1
Overall completion	~98%

Table 11: Implementation Completion Summary.

Statement: "We confirm that approximately 98% of the planned implementation has been completed as shown in the table above. The partially complete item (In-app downloadable final report) is a non-critical enhancement; all core audit functionalities are fully operational."

5.6 PDF Processing Architecture

The system uses a hybrid extraction architecture that prioritizes efficiency by avoiding unnecessary OCR operations:

Phase 1 — pdfplumber Extraction (Primary)

The system first attempts to extract invoice data directly using pdfplumber. This method is used for digital PDFs containing selectable text. Extracted fields include: invoice number, invoice date, VAT, total before VAT, grand total, and discount. If extraction confidence is sufficient, the system proceeds directly to matching. In the real-world test, 415 out of 438 invoices (94.7%) were processed via pdfplumber.

Phase 2 — OCR Fallback (Secondary)

If pdfplumber extraction fails or confidence is low, the system automatically switches to OCR. The OCR workflow: (1) convert PDF pages to images via pdf2image, (2) preprocess images using OpenCV (grayscale, thresholding), (3) extract text using Tesseract OCR (lang="ara+eng"), (4) parse financial values, (5) compare against Excel records. In testing, 23 invoices (5.3%) required OCR processing.

Extraction Method	Invoices Processed	Percentage
pdfplumber (Primary)	415	94.7%
Tesseract OCR (Fallback)	23	5.3%
Total	438	100%

Table 12: PDF Extraction Method Distribution.

5.7 Risk Classification System

The system calculates an overall audit score based on detected issues across all modules. The scoring system penalizes VAT errors, duplicate invoices, missing invoice sequences, AR reconciliation gaps, AP reconciliation gaps, and OCR mismatches.

Score Range	Risk Level	Description
85 – 100	LOW	Minimal issues; audit data is reliable
65 – 84	MEDIUM	Some discrepancies requiring review
40 – 64	HIGH	Significant issues requiring immediate attention
0 – 39	CRITICAL	Major failures; financial data unreliable

Table 13: Audit Risk Classification Score Ranges.

In the real-world test, AuditAI produced an audit score of 80/100 (MEDIUM risk level), primarily due to AR/AP reconciliation exceptions and 15 partial invoice matches.

Chapter 6: Testing

6.1 Test Plan and Approach

The testing strategy followed a structured bottom-up approach, starting from individual functions and progressing to full end-to-end system validation. Testing was conducted across three phases:

Phase	Level	Description
Phase 1	Unit Testing	Each function tested independently with isolated inputs
Phase 2	Integration Testing	All four modules tested together as a complete pipeline
Phase 3	System Testing	End-to-end test with full real company dataset

Table 14: Testing Phases Overview.

Testing Environment

- Platform: Google Colab (Python 3.10)
- Dataset: Real company data — ~500 PDF invoices (438 processed), 436 sales invoices, 37 AR customers, 12 AP suppliers
- Supporting files: AR aging, AP aging, receipt vouchers, payment vouchers
- Tools: Pandas, OpenPyXL, Python assertions, manual verification in Excel
- Dashboard: HTML/JavaScript browser-based validation

6.2 Types of Testing Performed

Test Type	Description	Tools Used	Status
Unit Testing	Each core function tested independently	Python assertions	Complete
Integration Testing	All four modules connected and tested as pipeline	Python, manual	Complete
Column Detection Testing	Smart Detector tested with Arabic/English/mixed names	Python assertions	Complete
Boundary & Edge Case Testing	Empty cells, zero values, large datasets, mixed types	Python, manual	Complete
Mathematical Verification	All formulas cross-verified against Excel calculations	Microsoft Excel	Complete
OCR Validation Testing	438 PDF invoices processed via pdfplumber (415) and OCR fallback (23)	pdfplumber, Tesseract, Python	Complete
Dashboard Output Testing	Web dashboard tested in browser for rendering and accuracy	Browser (Chrome)	Complete
Regression Testing	System re-run on same data to verify deterministic output	Python	Complete

Table 15: Types of Testing Performed.

6.3 Test Cases

The following test cases were conducted to verify the correctness, reliability, and consistency of the AuditAI system. The tests covered invoice validation, VAT verification, sequence checking, duplicate detection, and AR reconciliation scenarios to ensure that the system produces accurate audit results under different financial conditions.

Part 1 & 2 — Invoice Audit Test Cases

TC#	Description	Input	Expected Output	Actual Output	Pass/Fail
TC01	Correct VAT calculation	Pre-tax: 9,376 → VAT: 1,406.40	No error flagged	No error flagged	Pass
TC02	Incorrect VAT calculation	Pre-tax: 1,000 → VAT: 200 (wrong)	Flagged as VAT error	Flagged as VAT error	Pass
TC03	Sequential invoices (no gaps)	3362 to 3797 continuous	0 missing numbers	0 missing	Pass
TC04	Gap in invoice sequence	3362, 3363, 3365 (gap at 3364)	3364 flagged missing	3364 flagged	Pass
TC05	Duplicate invoice number	Invoice 3362 appears twice	Flagged as duplicate	Flagged	Pass
TC06	Total consistency check	Pre-tax: 855, VAT: 128.25 → Net: 983.25	No error flagged	No error	Pass

Table 16: Invoice Audit Test Cases (TC01–TC06).

Part 3 — AR Reconciliation Test Cases

TC#	Scenario	Formula Values	Classification	Pass/Fail
TC07	Balanced customer	$0 + 500 - 500 = 0 \checkmark$	OK	Pass
TC08	Sales not reflected in AR	$\text{Sales} > 0, \text{AR_close} = 0$	Sales Not Reflected	Pass
TC09	Receipts without sales	$\text{Sales} = 0, \text{Receipts} > 0$	Receipts without Sales	Pass
TC10	Advance receipts	$\text{Receipts} > \text{Sales}$	Possible Advance Receipts	Pass
TC11	General mismatch	$\text{Difference} > 1 \text{ SAR}$	AR Mismatch	Pass

Table 17: AR Reconciliation Test Cases (TC07–TC11).

Column Detection Test Cases

The following test cases were conducted to evaluate the accuracy and flexibility of the Smart Column Detector module. Different Arabic and English column naming variations were tested to ensure that the system can correctly identify and map financial fields across multiple file formats and naming conventions.

Standard Field	Test Name A	Test Name B	Result
vat	الضريبة	ض.ق.م	Detected correctly
invoice_code	كود الفاتورة	رقم الفاتورة	Detected correctly
total	الإجمالي	الإجمالي ق الضريبة	Detected correctly
grand_total	Grand Total	الإجمالي الكلي	Detected correctly
customer_code	الكود	كود العميل	Detected correctly

Table 18: Smart Column Detector Test Cases.

6.4 Verification Methods

- 6.4.1 Mathematical Verification:** All audit calculations and reconciliation formulas were manually verified in Microsoft Excel using the same financial datasets. The results matched the Python audit engine outputs with no calculation differences detected.
- 6.4.2 Cross-Run Consistency:** The system was executed multiple times using the same datasets to verify result stability and consistency. All generated outputs remained identical across runs, confirming reliable and deterministic system behavior.
- 6.4.3 Peer Review Testing:** Each module was reviewed and tested by different team members before full system integration to ensure correctness, detect potential issues, and improve overall reliability.

6.5 Conclusion of Testing Phase

The testing phase confirmed that the AuditAI system performs accurately and reliably across all four audit modules. The system successfully processed 438 PDF invoices with a success rate and a 96.6% match rate (24 full matches, 399 derived matches, 15 partial matches). VAT validation, invoice sequence, and duplicate detection returned zero errors across all invoices. AR reconciliation covered 37 customers (6 exceptions, 83.8% match rate) and AP reconciliation covered 12 suppliers (8 exceptions, 33.3% match rate — primarily due to incomplete data in the source dataset). The overall audit score was 80/100 (MEDIUM risk). The system is confirmed ready for production use.

Chapter 7: Results and Statistics

7.1 Audit Performance Results

AuditAI evaluation results using a complete real-company dataset are summarized below:

Overall System Performance

Metric	Score	Notes
Overall Audit Score	80 / 100	MEDIUM risk level
Invoice Match Rate	96.6%	423/438 invoices matched (full + derived)
VAT Errors	0	All 436 sales invoices passed VAT validation
Duplicate Invoices	0	No duplicates detected in sales or purchases
Missing Invoice Sequences	0	No gaps found in invoice numbering
AR Match Rate	83.8%	31/37 customers reconciled; 6 exceptions
AP Match Rate	33.3%	4/12 suppliers reconciled; 8 exceptions (incomplete source data)
Processing Time	18–32 minutes	~4.9 seconds per invoice (hybrid extraction)

Table 19: Overall AuditAI System Performance Metrics.

Audit Results by Module

Audit Module	Records Checked	Issues Found	Status
Sales — Invoice Sequence	436 invoices	0 gaps, 0 duplicates	Clean
Sales — VAT Validation	436 invoices	0 errors	Clean
Sales — Total Consistency	436 invoices	0 errors	Clean
Purchases — Sequence	Purchase orders	0 gaps, 0 duplicates	Clean
Purchases — VAT Validation	Purchase orders	0 errors	Clean
AR Reconciliation	37 customers	6 exceptions (total gap: 56,445 SAR)	MEDIUM risk — review required
AP Reconciliation	12 suppliers	8 exceptions (total gap: 579,779 SAR)	HIGH risk — incomplete source data
PDF Extraction (pdfplumber)	415 invoices	415 successful extractions	Clean
OCR Fallback (Tesseract)	23 invoices	23 successful extractions	Clean
Invoice Matching (Total)	438 invoices	15 partial matches, 0 errors	96.6% match rate

Table 20: Audit Results by Module.

Note: The high AP exception rate and gap values were primarily caused by incomplete supplier data in the provided dataset (missing opening/closing balances, incomplete transaction histories). Additionally, purchase invoice PDF matching was not performed as the company did not provide purchase invoice PDFs — OCR matching was applied to sales invoices only. These limitations do not reflect system failures; the AP reconciliation logic was fully implemented and correctly executed.

7.2 PDF Invoice Matching Results

Table 14 summarizes the PDF invoice matching and validation results generated by the AuditAI system. The table shows how invoices were classified after comparing extracted PDF data against Excel financial records using the hybrid extraction and matching process.

Match Category	Count	Description
Full Match	24	All extracted fields match Excel records exactly
Derived Match	399	Match confirmed; some values mathematically derived
Partial Match	15	Invoice exists but one or more fields differ
Not Found in Excel	0	Invoice exists in PDF but not in Excel
Error	0	Extraction or processing error
Total Processed	438	96.6% effective match rate (full + derived)

Table 21: Audit Results by Module.

7.3 Financial KPI Summary

Table 15 summarizes the key financial KPIs generated by the AuditAI system after processing and analyzing the company's financial datasets. These indicators provide a high-level overview of audit performance, reconciliation results, invoice validation accuracy, and detected financial gaps.

KPI	Result
Total Sales	4,610,000 SAR
Total Purchases	4,070,000 SAR
VAT Errors	0
Duplicate Invoices	0
Missing Invoices	0
PDF Match Rate	96.6%
AR Total Gap	56,445 SAR
AP Total Gap	579,779 SAR
Overall Audit Score	80 / 100 (MEDIUM)

Table 22: Financial KPI Summary.

7.4 Risk Classification

Table 16 presents the risk classification results generated by the AuditAI system based on detected discrepancies, reconciliation gaps, and invoice validation outcomes. Each audit area was evaluated and assigned a risk level to help auditors prioritize investigation and review activities.

Risk Area	Level	Details
Sales Invoices	LOW	0 errors across VAT, sequences, duplicates, totals
Purchase Invoices	LOW	0 errors across all validation checks
AR Reconciliation	MEDIUM	6 exceptions; manageable gaps under investigation
OCR Partial Matches	MEDIUM	15 invoices with minor field discrepancies
AP Reconciliation	HIGH	8 exceptions; largest gap (579,779 SAR) — data completeness issue

Table 23: Risk Classification by Audit Area.

Conclusion and Future Work

Summary of Achievements

Table 17 summarizes the main objectives achieved during the development of AuditAI. The results demonstrate the system's ability to automate financial audit tasks, validate invoices, perform reconciliation processes, support multilingual financial data, and provide interactive audit analysis through a web-based platform.

Original Aim	Achieved?	Evidence
Automate AR/AP financial reconciliation	Yes	37 AR customers + 12 AP suppliers reconciled
Detect invoice anomalies (VAT, sequence, duplicates)	Yes	0 errors found across all validation checks
Support Arabic and English data formats universally	Yes	Smart Column Detector; OCR lang="ara+eng"
Hybrid PDF invoice validation (pdfplumber + OCR)	Yes	438 invoices: 415 pdfplumber + 23 OCR
96%+ invoice match rate	Yes	96.6% match rate (423/438 invoices)
Interactive web dashboard for auditors	Yes	Section D.3, Figure 5
Complete web application with authentication	Yes	Section D.3, Figures 1–4
Process large invoice datasets	Yes	~500 invoices in 18–32 min (4.9 sec/invoice)
Risk classification system	Yes	Score: 80/100 MEDIUM; 4-level risk scale

Table 24: Summary of Project Achievements.

Challenges Faced

Challenge 1: Multi-Source Data Inconsistency

Financial files from different sources had inconsistent column names, formats, and data types. To address this, the system uses a Smart Column Detector and preprocessing logic to map required columns, standardize formats, handle missing values, and prepare the files for audit processing.

Challenge 2: Real-World Accounting Complexity

Some accounting cases are not simple errors, such as advance payments, partial settlements, and timing differences. The system addresses this by using rule-based classification logic to group discrepancies into meaningful audit categories, helping auditors understand which cases require review.

Challenge 3: PDF Invoice Matching Limitations

Invoice PDFs may contain different formats, scanned pages, or Arabic text, which makes extraction difficult. The system uses a hybrid approach: pdfplumber is applied first for text-based PDFs, while OCR is used as a fallback when direct extraction is not sufficient. This improves invoice matching and validation accuracy.

Future Work

- **Advanced AI Anomaly Detection:** Integrate machine learning models (e.g., Isolation Forest, Autoencoder) for unsupervised anomaly detection in transaction patterns beyond rule-based logic.
- **Cloud Integration:** Connect AuditAI to cloud-based accounting platforms (e.g., SAP, Oracle Financials, Zoho Books) for direct data ingestion without manual file uploads.
- **Automated Report Generation:** Complete the in-app audit report generation feature with professional PDF output formatted according to international auditing standards (ISA/IFRS).
- **NLP-Based Audit Recommendations:** Integrate NLP models to generate intelligent audit insights, recommendations, and explanations automatically within the final audit report.
- **Extended Language Support:** Expand the Smart Column Detector using NLP techniques to support additional languages, naming conventions, and accounting system formats used across GCC markets.

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Appendices

Appendix A: Final Checklist

Item	Done?
Abstract (200-300 words, English & Arabic, no citations)	✓
Section A: Summary of previous work (within 2 pages)	✓
Section B: Methods and approaches (development phases)	✓
Section C: Activities (completed + not completed + why)	✓
Section D.1: Tools and technologies table with justification	✓
Section D.2: Interface type described and justified	✓
Section D.3: Screenshots with figure numbers and captions	✓
Section D.4: User manual for laymen (simple steps)	✓
Section D.5: Implementation progress table (70%+ Part I, 98%+ Final)	✓
Section E: Testing (plan, types, test cases, conclusion)	✓
Section F: Results and statistics	✓
Section G: Conclusion and future work	✓
Section H: References (APA + extra bibliography)	✓
Section I: Appendices	✓
File named: CPCS499_Group_AuditAI_Final	✓

Table 25: Final Report Checklist.

Appendix B: System Architecture Overview

The AuditAI system follows a modular three-tier architecture:

- **Presentation Layer:** A web-based user interface that allows auditors to manage clients, upload financial documents, run audits, and review results through dashboards and visual reports.
- **Business Logic Layer:** A Python-based audit engine responsible for processing financial data, performing reconciliation and validation checks, detecting discrepancies, matching invoices, and generating audit outputs. This layer includes the Sales Audit, Purchases Audit, AR/AP Reconciliation modules, Smart Column Detector, and PDF validation components.
- **Data Layer:** A flexible file-based data system that supports Excel and PDF financial documents. The system generates structured outputs for dashboards and downloadable audit reports. Supabase is used for authentication, user management, and backend data handling.

Appendix C: Audit Engine Module Summary

Module	Input	Checks Performed	Output
Sales Audit	Sales invoices Excel	VAT validation, invoice sequence gaps, duplicate detection, total consistency checks	436 invoices processed with 0 validation errors
Purchases Audit	Purchase orders Excel	Same validation checks as Sales Audit	Validation completed; purchase PDFs not available
AR Reconciliation	AR opening/closing balance, sales, receipts	Reconciliation of opening balance, sales, receipts, and closing balance	37 customers analyzed; 6 exceptions detected
AP Reconciliation	AP opening/closing balance, purchases, payments	Reconciliation of opening balance, purchases, payments, and closing balance	12 suppliers analyzed; 8 exceptions detected
PDF Validation	Sales invoice PDFs + Excel data	PDF extraction, OCR fallback, and invoice field matching	438 invoices processed with 96.6% match rate

Table 26: Audit Engine Module Summary.